

REVIEW

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Performance evaluation of finger-worn devices for sleep stage classification and sleep apnea detection: a systematic review and meta-analysis

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Abstract

Background Recently, sleep wearables have experienced a boom, particularly devices worn on the finger, such as smart rings, which have garnered growing attention. However, there currently exists no review that comprehensively and systematically evaluates the accuracy of these devices.

Main body This scoping review evaluates 11 finger-worn devices against polysomnography (PSG) across 28 articles from inception to April, 2025, assessing sleep staging (primarily healthy participants) and obstructive sleep apnea (OSA) detection (clinical cohorts) via heterogeneous sensors and algorithms. Current evidence in our study demonstrates efficacy in sleep/wake classification (pooled accuracy = 87%, 95%CI = 86%~89%) and severe OSA screening, but reveals persistent challenges in multi-stage sleep analysis (pooled accuracy for light/deep/rapid eye movement (REM) sleep = 0.65, 0.81, 0.74, respectively) and mild OSA detection. All devices, except one, demonstrated higher accuracy at an apnea-hypopnea index (AHI) threshold of 30 than at thresholds of 5 or 15.

Conclusions Finger-worn devices may be used for longitudinal sleep/wake tracking in healthy adults or as a triage tool for severe OSA, but should not replace PSG for diagnosis of mild OSA or detailed sleep architecture assessment. Achieving PSG-comparable accuracy in multi-stage sleep analysis and mild OSA detection requires addressing algorithmic transparency, population diversity, and standardized methodologies.

Keywords Finger-worn devices, Polysomnography, Sleep stage, Sleep apnea

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Background

Sleep, a modifiable determinant of health, plays a pivotal role in maintaining physiological homeostasis and cognitive function. Robust evidence links adequate sleep duration and quality to metabolic regulation, immune competence, and cardiovascular health [1, 2], while sleep deficiencies are associated with impaired neurobehavioral performance, mood disorders, and increased risk of chronic disease [3]. Sleep can be assessed subjectively by using questionnaires or objectively through measurement of physiological sleep parameters. Although polysomnography (PSG) remains the gold standard for evaluating sleep architecture and respiratory events, its clinical utility is limited by technical complexity, high cost, accessibility constraints, and patient discomfort from multi-sensor attachments in unfamiliar environments [4]. These limitations necessitate multi-night recordings to counter the “first-night effect,” yet such approach remains impractical for longitudinal or community-based studies.

The advent of consumer-grade wearable technologies has revolutionized sleep monitoring, offering scalable, cost-effective alternatives for both clinical and ambulatory settings, including wrist-worn devices, rings, and headbands [5]. Wrist-worn actigraphy, historically the dominant form of wearable sleep tracking, utilizes accelerometry to infer sleep-wake patterns but exhibits limited sensitivity to subtle movements characteristic of light sleep or transient arousals. Recent innovations in miniaturized multi-sensor systems have shifted focus toward alternative anatomical placements, with finger-worn devices emerging as a promising platform. The wrist actigraphy devices typically only detect significant body movements involving the forearm, while a device placed on the finger may be able to detect much smaller movements from the hand and finger that occur during wakefulness and light stages of sleep. Therefore, this location provides better signal quality and is less affected by noise, resulting in more accurate detection of sleep stages. Traditional pulse oximeters use photoplethysmography (PPG) sensors, which are also commonly placed on fingertips, toes, or earlobes. These devices capitalize on the finger’s rich vascularity and fine motor activity, enabling simultaneous acquisition of PPG, accelerometry, and pulse oximetry data. Such multimodal integration enhances detection of sleep stages and respiratory events, particularly obstructive sleep apnea (OSA), a globally prevalent disorder with significant cardiometabolic morbidity [6].

Early-generation finger-worn devices demonstrated modest accuracy in sleep staging. For instance, the Oura Ring Gen1 achieved 65% agreement with PSG for light sleep, 51% for deep sleep, and 61% for REM sleep [7]. Subsequent iterations, such as the Oura Ring Gen2, reported marked improvements (80% light sleep, 91%

deep/REM sleep accuracy) through advanced algorithms and sensor fusion [8]. The two-stage classification (wake/sleep) also demonstrated higher yet inconsistent accuracy, ranging from 0.85 [9] to 0.96 [8], depending on specific devices used. Furthermore, the identification of OSA severity varied across different devices, such as the Belun Ring [10], Circul Ring [11], and RingConn Smart Ring [12]. These disparities underscore the need for rigorous, standardized review for the performance of finger-worn technologies against PSG benchmarks.

Despite expanding research interest, no systematic synthesis exists that evaluates finger-worn devices in sleep staging and obstructive sleep apnea (OSA) detection. This review addresses this critical gap by comprehensively appraising recent technological advancements, examining their accuracy against polysomnography, methodological rigor of validation studies, and clinical implications.

Methods

The study was registered on PROSPERO prospectively (ID CRD42024605972) and was performed in compliance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses).

Search strategy

We performed a systematic search including the databases of PubMed, Web of Science, and Scopus to pinpoint relevant papers published to 8 April 2025. The search was conducted using both MeSH (medical subjects heading terms) and free-text words such as ‘Finger-worn device,’ ‘Wearable ring,’ ‘Sleep stages,’ ‘Sleep Apnea Syndromes,’ and ‘Detect,’ with appropriate use of Boolean operators (AND, OR) to combine search terms. We limited the search to English language articles. Additionally, the reference lists of relevant articles were searched to identify any additional articles. We also reviewed the official websites of major smart ring manufacturers for any available validation study reports and examined clinical trial for completed but potentially unpublished studies. The complete search words and search strategy are provided in Supplementary Table 1.

Selection process

Based on the search, the study selection process was conducted by two independent investigators (YXJ and JWX). Any disagreements were addressed by a third reviewer (JW). Supplementary Table 2 presents the inclusion and exclusion criteria for study selection. The inclusion criteria were the publications that involved population using any finger-worn device, that reported the performance of detecting sleep stages and sleep apnea, and that incorporated PSG as reference. Letters, comments, conference abstracts and review articles were excluded. If the study

was published repetitively, the most recent study or the study with larger sample size was selected. Exclusion criteria also included studies that utilized non-ring devices (e.g., Watch, Fitbit, or WHOOP). The study selection process is shown in Fig. 1.

Data extraction and analysis

Data extraction involved study characteristics (e.g., first author, year of publication, location, research setting, and duration), participants demographics (e.g., sample size, mean age, sex, and health status), and device-specific attributes (e.g., type, algorithm, feature selection, and epoch length). Validity refers to the extent to which a measurement tool reflects the actual characteristic it is intended to assess [13]. In this study, we mainly focused on criterion validity, which examines criterion-related validity by benchmarking finger-worn devices against the gold-standard polysomnography (PSG). Therefore, we documented the diagnostic performance outcomes in terms of accuracy, sensitivity, specificity, balanced accuracy and Cohen's kappa. The detailed indices are as follows:

Accuracy: the proportion of all test results (both correct and incorrect) that are correctly classified.

$$\text{Accuracy} = \frac{\text{True position} + \text{True negative}}{\text{True position} + \text{True negative} + \text{False positive} + \text{False negative}}$$

Sensitivity: the ability of a test to correctly identify individuals with the condition (true positives).

$$\text{Sensitivity} = \frac{\text{True position}}{\text{True position} + \text{False negative}}$$

Specificity: the ability of a test to correctly exclude individuals without the condition (true negatives).

$$\text{Specificity} = \frac{\text{True negative}}{\text{True negative} + \text{False positive}}$$

Balanced accuracy: indicates the mean of wake epochs correctly classified and sleep epochs correctly classified, calculated as the mean of sensitivity and specificity.

Cohen's kappa (κ): is calculated to evaluate the agreement values relative to that which could be expected due to chance.

We conducted a meta-analysis to provide summary measures for sleep stages. Given the heterogeneity across these studies, we chose random effects models in the meta-analysis. The analysis was conducted by

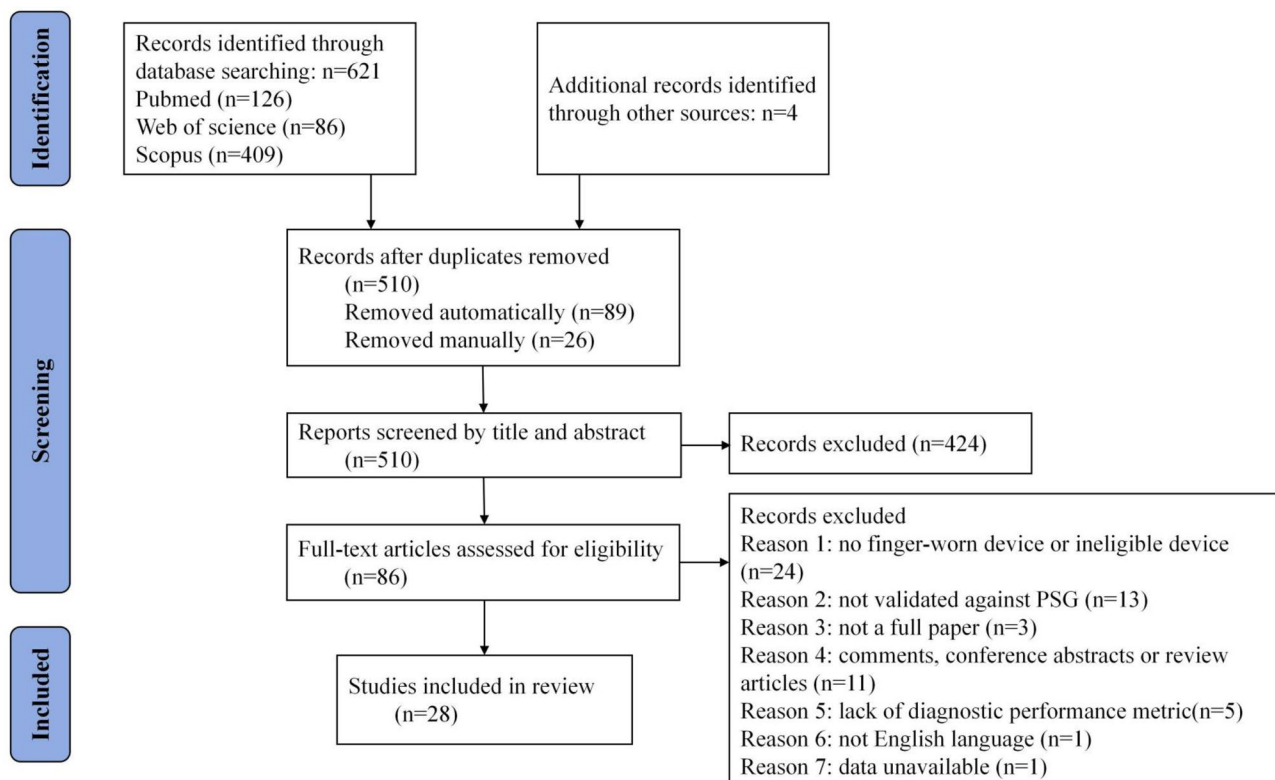


Fig. 1 PRISMA flow diagram

implementing R statistical software (version 4.3.2) and the “metagen” R packages (version 4.3.2).

Evaluation of quality

The Quality Assessment of Diagnostic Accuracy Studies-2 (QUADAS-2) tool was used to evaluate the quality of the primary studies [14]. The risk of bias included four main domains: participants, index test, reference standard, and flow and timing. In every domain, risk was classified as ‘high’ if the criteria did not meet at least one of the signaling criteria, ‘unclear’ if information was unavailable for one or more criteria, and ‘low’ if all signaling criteria were fully satisfied. If a study is judged as “low” on all domains relating to bias then it is appropriate to have an overall judgment of “low risk of bias” for that study. If a study is judged “high” or “unclear” on one or more domains then it may be judged “at risk of bias”. Quality assessment was performed by two independent investigators (YXJ and JWX), with any differences resolved through discussion and consensus.

Results

Study characteristics

We involved 28 articles including 2426 participants in the review, with a subset analysis of 14 articles [7–9, 15–25] explored the performance of finger-worn devices in sleep stages, and 13 articles [10–12, 26–35] revealed the detection on sleep apnea. One article [36] examined the above two aspects (Tables 1 and 2; Fig. 2). Supplementary Table 3 shows the basic characteristics of included studies across 28 articles. The systematic analysis encompassed researches spanning 2018–2025, with 20 studies emerging in or after 2022. In terms of participant demographics, the sleep stage studies predominantly involved adults aged 18–50 years (12/15) and participants with normal weight (8/15). In contrast, the sleep apnea studies more frequently included participants over 50 years old (9/14) and those who were overweight or obese (12/14). For disease severity in apnea-focused studies, the majority of studies simultaneously reported the AHI of 5, 15, 30 (11/14). Notably, only three studies targeted children and adolescents in the above two aspects. 15 studies employed 30-second epochs for analysis.

Figure 2 illustrates the distribution of included participants by health status. In studies utilizing finger-worn devices for sleep staging, 86% (12/14) predominantly involved healthy participants. Conversely, for OSA assessment, 77% (10/13) of the studies focused on clinical cohorts. Specifically, among the OSA-focused studies, eight involved participants with suspected sleep apnea, two focused on individuals with cardiovascular disease (CVD), and only two included healthy participants.

Algorithms for finger-worn devices

Eight (73%) finger-worn devices predominantly employed proprietary algorithms (Tables 1 and 2). Oura Ring’s algorithms performed iterative advancements: the first generation model combined the physiological signal with a gradient boosting classifier to evaluate classification performance [15]. The second generation model employed LightGBM with DART optimization [8]. The third generation OSSA 2.0 integrated multimodal data from a 3D accelerometer, seven skin temperature sensors, and infrared PPG to enhance staging precision [24]. Hannah et al. developed THIM sleep tracking algorithm scored the sleep or wakefulness every 30s, which incorporated a smoothing function [9]. The Happy Ring platform implemented a dual algorithmic framework: a “generalized” algorithm adopted leave-one-out cross-validation (LOOCV) for all users, and a “personalized” model utilized non-overlapping time window and a five-fold cross-validation to preserve sleep pattern continuity [22]. Belun ring integrated a medical-grade ring pulse oximeter with a proprietary neural-network-based artificial intelligence algorithm [10, 36]. Circul Ring automatically processed embedded rPPG and accelerometer (ACC) signals [11].

Types of devices and sensors

This comprehensive review critically evaluated 11 advanced wearable rings, with a particular focus on their specialized applications (Figs. 3 and Supplementary Fig. 1): the Oura ring, SleepOn, THIM, and Happy ring were meticulously examined for their capabilities in sleep stage monitoring and analysis, while the NightOwl, Biologix™, SleepImage, RingConn smart ring and Wellue O2 ring were rigorously evaluated for efficacy in sleep apnea detection. The Circul ring and Belun ring enabled assessment of both sleep stages and sleep apnea.

The sensors of finger-worn device were shown in Fig. 3 and Supplementary Fig. 2. Eight devices utilized multi-sensor to collect physiological, movement, and other health-related metrics. Specifically, four devices [7, 11, 28, 35] utilized ACC and PPG to measure body signals. Three smart rings [8, 24, 34] incorporated data from ACC, PPG, and temperature sensors. Furthermore, Happy Ring [22] not only utilized the above three sensors, but also relied on electrodermal activity (EDA) sensors to facilitate multidimensional analyses of physical activity, peripheral blood flow, and distal thermoregulation. Two finger-worn devices sensors were unclear.

Performance in sleep stage detection versus PSG

For wake/sleep classification, as shown in Fig. 4, a meta-analysis of seven studies revealed a pooled accuracy of 0.87 (0.86,0.89). Specifically, the Oura Ring Generation 2 with Gen 3 algorithm reported by Ghorbani et al. demonstrated the highest accuracy at 0.93, while the Circul

Table 1 Overview of articles related to sleep stage

Author	Location	Number of participants	Age	Health status	Device	Used sensors	Environment	Time	Algorithm	Feature selection	Epoch length	Group	2 stages accuracy (wake/sleep)	4 stages accuracy deep/REM	Wake sleep	Light sleep	Deep sleep	REM
de Zambotti et al. 2019 [7]	America	41 (13 female)	17.2±2.4 years	Healthy	Oura Ring Generation 1	ACC, PPG	Laboratory	2 days	Proprietary algorithm (first version of Ouraring algorithm)	N/A	30s	N/A	N/A	N/A	N/A	0.65 (0.60, 0.69)	0.51 (0.43, 0.59)	0.61 (0.54, 0.69)
Roberts et al. 2020 [15]	America	8 (5 female)	40.8±4.8 years	Healthy	Oura Ring Generation 1	ACC, PPG	Laboratory	4 days	Proprietary algorithm (Sleep-wake classification and motion algorithms)	Mean IBI, SD of IBI, motion seconds	30s	N/A	0.90 (0.86, 0.94)	N/A	N/A	N/A	N/A	N/A
Chee et al. 2021 [16]	Singapore	53 (28 female)	15.0–19.0 years	Healthy	Oura Ring Generation 2	ACC, PPG	Semi-laboratory	15 days	Proprietary algorithm	N/A	30s	Split sleep group	0.89 (0.87, 0.91)	N/A	N/A	0.52 (0.50, 0.54)	0.79 (0.74, 0.84)	0.53 (0.45, 0.61)
												Continuous group	0.89 (0.87, 0.91)	N/A	N/A	0.52 (0.49, 0.55)	0.83 (0.79, 0.87)	0.51 (0.44, 0.58)
												Adaptation group	0.89 (0.88, 0.90)	N/A	N/A	0.52 (0.50, 0.54)	0.79 (0.76, 0.82)	0.53 (0.48, 0.58)

Table 1 (continued)

Author	Location	Number of participants	Age	Health status	Device	Used sensors	Environment	Time	Algorithm	Feature selection	Epoch length	Group	2 stages accuracy (wake/sleep)	4 stages accuracy (wake/deep/REM)	Wake sleep	Light sleep	Deep sleep	REM
Al-tini and Kinnunen 2021 [8]	Singapore, Finland and America	Singapore:59 (30 female); Finland:19 (11 female); USA:40 (24 female)	30.1 ± 16.4 years	Healthy	Oura Ring Generation 2	ACC, PPG, temperature	Laboratory or home	Singapore: 1–8 days; Finland: 1–2 days; USA: 1–7 days	Proprietary algorithm	Mean, maximum, inter-quartile, mean amplitude deviation, arm angle, and HRV features	30s	N/A	0.96	N/A	0.96 (0.96, 0.97)	0.80 (0.80, 0.81)	0.91 (0.90, 0.91)	0.91 (0.90, 0.91)
Ghorbani et al. 2022 [17]	Singapore	58 (26 female)	37.1 ± 13.0 years	Healthy	Oura Ring Generation 2	ACC, PPG, temperature	Home	3 days	Gen 3 algorithm, Gen 2 algorithm	N/A	30s	Gen2	0.90 (0.88, 0.91)	0.62 (0.60, 0.64)	N/A	N/A	N/A	N/A
Miller et al. 2022 [18]	Australia	53 (26 female)	25.4 ± 5.9 years	Healthy	Oura Ring Generation 2	ACC, PPG	Laboratory	1 day	Proprietary algorithm	Sleep time, and HRV features	5 min	N/A	0.89	0.61	N/A	N/A	N/A	N/A
Taeyoung Lee et al. 2023 [19]	Korea	75 (36 female)	43.6 ± 14.1 years	Subjective sleep discomfort	Oura Ring Generation 3	ACC, PPG	Hospital	1 day	Proprietary algorithm	N/A	30s	N/A	N/A	0.54	0.82	0.62	0.80	0.85

Table 1 (continued)

Author	Location	Number of participants	Age	Health status	Device	Used sensors	Environment	Time	Algorithm	Feature selection	Epoch length	Group	2 stages accuracy (wake/sleep)	4 stages accuracy (wake/deep/REM)	Wake sleep	Light sleep	Deep sleep	REM
Ju Lynn Ong et al. 2024 [24]	Singapore	60 (34 female)	38.5 ± 15.1 years	Healthy	Oura Ring Generation 3	ACC, PPG, temperature	Laboratory	1 day	Oura Sleep Staging Algorithm 2.0 (OSSA 2.0)	N/A	30s	N/A	0.91 (0.89, 0.93)	0.74 (0.69, 0.79)	0.76 (0.74, 0.78)	0.74 (0.68, 0.80)	0.82 (0.78, 0.86)	
Thomas Svensson et al. 2024 [25]	Japan	96 (49 female)	41.9 ± 13.8 years	Healthy	Oura Ring Generation 3	ACC, PPG, NTC	Hotel	1-3 days	Oura Sleep Staging Algorithm 2.0 (OSSA 2.0)	ACC, HR, HRV features, and temperature	30s	Non-dominant hand	0.92 (0.92, 0.92)	N/A	N/A	0.76 (0.75, 0.76)	0.89 (0.89, 0.89)	0.91 (0.91, 0.91)
Rebecca Robbins et al. 2024 [20]	America	35 (20 female)	20–50 years	Healthy	Oura Ring Generation 3		Hospital	1 day	Oura Sleep Staging Algorithm 2.0	N/A	30s	N/A	0.92 (0.92, 0.92)	0.76 (0.76, 0.76)	N/A	N/A	0.89 (0.89, 0.89)	0.91 (0.91, 0.91)
Sebastian Herberger et al. 2025 [21]	Europe	45 (20 female)	54.6 ± 12.2 years	Unhealthy	Oura Ring Generation 3	ACC, PPG, NTC	Laboratory	1 day	N/A	Blood oxygen, HR, HRV, respiration, skin temperature, and movement	5 min	Oura ring	0.85 (0.82, 0.88)	0.53	N/A	N/A	N/A	N/A

Author	Location	Number of participants	Age	Health status	Device	Used sensors	Environment	Time	Algorithm	Feature selection	Epoch length	Group	2 stages accuracy (wake/sleep)	4 stages accuracy deep/REM	Wake sleep	Light sleep	Deep sleep	REM
Hannah Scott et al. 2021 [9]	Australia	Development group: 25 (15 female)	25.4 ± 6.4 years	Healthy	THIM (firmware v 1.0.3)	ACC	Laboratory	1 day	N/A	Blood oxygen, HR, HRV, ECG, blood pressure, and finger temperature	30s	Circulating	0.66 (0.62, 0.70)	0.35	N/A	N/A	N/A	N/A
					SleepOn	N/A	Laboratory	1 day	N/A	Blood oxygen, HR, and movement	60s	SleepOn	0.84 (0.81, 0.87)	0.50	N/A	N/A	N/A	N/A
										N/A		Development group	0.85 (0.82, 0.88)	N/A	N/A	N/A	N/A	N/A
												Test group	0.85 (0.82, 0.88)	N/A	N/A	N/A	N/A	N/A

Table 1 (continued)

Author	Location	Number of participants	Age	Health status	Device	Used sensors	Environment	Time	Algorithm	Feature selection	Epoch length	Group	2 stages accuracy (wake/sleep)	4 stages accuracy deep/REM	Wake sleep	Light sleep	Deep sleep	REM
Michael A. Grandner et al. 2023 [22]	America	36 (16 female)	23.4±3.8 years	Healthy	Happy Ring	ACC, PPG, EDA, temperature	Laboratory	3 days	Happy ring: proprietary algorithm (a "generalized" algorithm; a "personalized" algorithm)	Mean, SD, range, frequency of values and variables	30s	Personalized group	0.92 (0.91, 0.93)	N/A	N/A	0.81 (0.79, 0.83)	0.95 (0.94, 0.96)	0.92 (0.91, 0.93)
Zachary Strumpf et al. 2023 [36]	America	84 (42 female)	48.3±14.4 years	Confirmed OSA and suspect OSA	Belun Ring(BSP2)	ACC, PPG	Laboratory	1 day	DL-enabled BSP2 algorithms	SpO2, PR, and ACC signal	30s	N/A	N/A	N/A	0.88 (0.87, 0.88)	0.82 (0.81, 0.82)*	0.90 (0.90, 0.90)	0.90 (0.90, 0.90)
Adrian R. Wiloughby et al. 2024 [23]	Singapore	63 (36 female)	39.6±15.5 years	Healthy	Oura Ring-Generation 3	N/A	Laboratory	2 days	Oura Sleep Staging Algorithm 2.0	N/A	30s	N/A	N/A	N/A	Sleep period 1: 0.82	Inter-ruption period 1: 0.94	Sleep period 2: 1.00	Morning wake: 0.98

Note: *Indicates the non-rapid eye movement (NREM), including light sleep and deep sleep
Abbreviations: ACC=accelerometer, PPG=photoplethysmography, NTC=negative temperature coefficient, IBI=interbeat interval, SD=standard deviation, HRV=heart rate variability, HR=heart rate, SpO2= oxygen saturation, PR=pulse rate, EDA=electrodermal activity, REM=rapid eye movement, OSA=obstructive sleep apnea, N/A=not applicable

Table 2 Overview of articles related to sleep apnea

Author	Location	Number of participants	Age	Health status	Device	Used sensors	Environment	Algorithm	Feature selection	Epoch length	AHI/OSA cut off	Accuracy
Wenbo Gu et al. 2020 [26]	America	50 (23 female)	54.9 ± 16.3 years	Without major comorbidities or heart rate-affecting medications	Belun Ring	ACC, PPG	Laboratory	Neural network algorithm	SpO2, PR, HRV, ACC signal, and PPG waveform	30s	5	0.86
Eric Yeh et al. 2021 [10]	America	78 (51 female)	51.5 ± 13.4 years	On heart-rate-affecting medications or with non-arrhythmic conditions	Belun Ring (BSP1)	ACC, PPG	Laboratory	BSP proprietary OSA detection algorithm	SpO2, PR, HRV, ACC signal, and PPG waveform	5 min	15	0.86
											30	0.86
											5	0.56 (0.45,0.68)
Ou, Yi-Hui et al. 2023 [35]	Singapore	129 (15 female)	60.5 ± 8.4 years	With hypertension and high cardiovascular risk	Belun ring	ACC, PPG	Laboratory	N/A	N/A	N/A	15	0.81 (0.70,0.89)
											30	0.91 (0.82,0.96)
											5	0.84
Zachary Strumpf et al. 2023 [36]	America	84 (42 female)	48.3 ± 14.4 years	Confirmed OSA and suspected OSA	Belun Ring (BSP2)	ACC, PPG	Laboratory	DL-enabled BSP2 algorithms	SpO2, PR, and ACC signal	5 min	15	0.87
											30	0.92
											5	0.85 (0.75,0.92)
Nuttida Panichapat et al. 2024 [27]	Thailand	75 (21 female)	9.2 ± 2.8 years	Suspected OSA	Belun Ring	ACC, PPG	Hospital	Neural network algorithm	SpO2, HRV, ACC, and PPG	N/A	15	0.89 (0.81,0.95)
											30	0.91 (0.82,0.96)
											3	0.68
Frederik Massie et al. 2018 [28]	Belgium	101 (44 female)	53.0 ± 13.0 years	Suspected OSA	NightOwl	ACC, PPG	Laboratory	N/A	Actigraphy, SpO2, PAT, PR, and other features	N/A	5	0.52
											7	0.44
											Overall	0.81
Bart Van Pee et al. 2022 [29]	Belgium, America	167 (62 female)	56.0 ± 15.0 years	Suspected OSA	NightOwl	ACC, PPG	Laboratory	N/A	N/A	N/A	5	0.90 (0.86,0.95)
											15	0.86 (0.80,0.91)

Table 2 (continued)

Author	Location	Number of participants	Age	Health status	Device	Used sensors	Environment	Algorithm	Feature selection	Epoch length	AHI/OA cut off	Accuracy
Christopher J. Lyne et al. 2023 [30]	Australia	100 (43 female)	48.0 ± 11.8 years	Suspected OSA	NightOwl	ACC, PPG	Hospital	Proprietary algorithm	Actigraphy, PAT	N/A	30	0.91 (0.87,0.95) 0.64
											NightOwl reusable overall	0.68
											Moderate-to-severe OSA, ODM-ODI cut off = 12, 3% oxygen desaturation	0.90
George do Lago Pinheiro et al. 2020 [31]		304 (135 female)	55.3 ± 13.8 years	Suspected OSA	Biologix™	ACC	Laboratory	Auto-mated cloud algorithm	N/A	1 s		0.90
Rui Zhao et al. 2022 [11]	China	207 (62 female)	48.2 ± 14.7 years	Suspected sleep apnea	Circul Ring	ACC, PPG	Laboratory	Automated algorithm	SpO ₂ , PR, and ACC signal	1 s	5	0.86
Mi Lu et al. 2023 [32]	German	39 (10 female)	56.8 ± 15.0 years	Suspected sleep apnea	Sleepimage Ring	PPG	Laboratory	Cloud-based Sleepimage system	PR variability, respiration coherence, cross spectral power, and SpO ₂	N/A	15 30 5	0.76 0.91 1.00 (1.00,1.00)
Zetong Wu et al. 2024 [12]	America	58	54.0–93.0 years	Without clinical cardiovascular disease	RingConn smart ring	PPG	N/A	N/A	Trend components, periodic component, and SpO ₂	60 s	5	0.83 0.83

Table 2 (continued)

Author	Location	Number of participants	Age	Health status	Device	Used sensors	Environment	Algorithm	Feature selection	Epoch length	AHI/OSA cut off	Accuracy
Hongbo Guo et al. 2024 [34]	China	58 (17 female)	52.2 ± 12.9 years	N/A	RingConn smart ring	ACC, PPG, temperature	Hospital	N/A	SpO2, ACC, PPG, and temperature	60s	30 5 15 30	0.88 0.81 0.86 0.88
James Tisyakorn et al. 2024 [33]	Thailand	190 (89 female)	43.0 ± 16.0 years	Suspected OSA	Wellue O2 ring	N/A	Laboratory	N/A	N/A	20s	Moderate-to-severe OSA, 11% ODI cut off = 1.25	N/A

Abbreviations: ACC=accelerometer, PPG=photoplethysmography, OSA=obstructive sleep apnea, SpO2= oxygen saturation, PR=pulse rate, HRV=heart rate variability, PAT=peripheral arterial tone, ODM=overnight digital monitoring, ODI=oxygen desaturation index, N/A =not applicable

ring reported by Sebastian et al. exhibited the lowest accuracy at 0.66.

For four-stage sleep classification, a meta-analysis of five studies (Fig. 5) yielded pooled accuracy of 0.65 (0.56,0.74) for light sleep, 0.81 (0.77,0.85) for deep sleep, and 0.74 (0.68,0.79) for REM sleep detection. Specifically, Michael A et al. reported that the Happy Ring with personalized algorithm, demonstrated the highest accuracy rates: 0.81 (0.79,0.83) for light sleep, 0.95 (0.94,0.96) for deep sleep, and 0.92 (0.91,0.93) for REM sleep. In contrast, the Oura Ring Generation 2 reported by Chee et al. showed the lowest light (0.52) and REM (0.51) sleep accuracy, while Generation 1 reported by de Zambotti showed the lowest deep sleep accuracy (0.51). The leave-one-out analysis confirmed the robustness of these results (Supplementary Figs. 3 and 4). Notably, only the Belun ring performed three-stage classification, which reported accuracy of 0.88 for wake, 0.82 for NREM, and 0.90 for REM [36] (Table 1).

In two-stage sleep classification compared with PSG, finger-worn devices demonstrated sensitivity ranging from 0.74 to 0.97, specificity ranging from 0.35 to 0.98, and balanced accuracy ranging from 0.55 to 0.90. For four sleep stages, sensitivity ranges were 0.45–0.82 for light sleep, 0.17–0.89 for deep sleep, and 0.14–0.89 for REM sleep, with corresponding specificity values of 0.64–0.91, 0.80–0.95, 0.87–0.94 across finger-worn devices. Reported Cohen’s kappa ranged from 0.06 to 0.70, with a considerable number of studies not reporting the metric. Details are shown in Supplementary Tables 4–6.

Performance in sleep apnea detection versus PSG

Figure 6 shows that finger-worn devices predominantly utilize AHI thresholds of 5, 15, and 30 events/hour for severity classification in adults. At the AHI = 5 events/hour, the SleepImage Ring demonstrated the highest accuracy (1.00), whereas the Belun Ring in Eric et al.’s study exhibited the lowest accuracy (0.56). For using an AHI threshold of at least 15 events/hour, the Belun Ring reported by Zachary et al. achieved an accuracy of 0.89, but Circul Ring reported 0.76. At the severe threshold (AHI = 30 events/hour), the Belun Ring reported by Ou et al. maintained high accuracy (0.92), while the RingConn smart ring reported by Wenbo et al. showed the lowest performance (0.86). Notably, all of devices displayed higher accuracy at AHI thresholds of 30 compared to the cutoff of 5 and 15, except for the SleepImage Ring. Besides, Nuttida et al. [27] highlighted declining accuracy for Belun ring at low thresholds in children: 0.68, 0.52, 0.44 at AHI = 3, 5, and 7 events/hour.

Beyond overall accuracy, the sensitivity, specificity, and Cohen’s kappa of finger-worn devices for OSA detection across various AHI thresholds are detailed in Supplementary Table 7. At the threshold of AHI=5, sensitivity

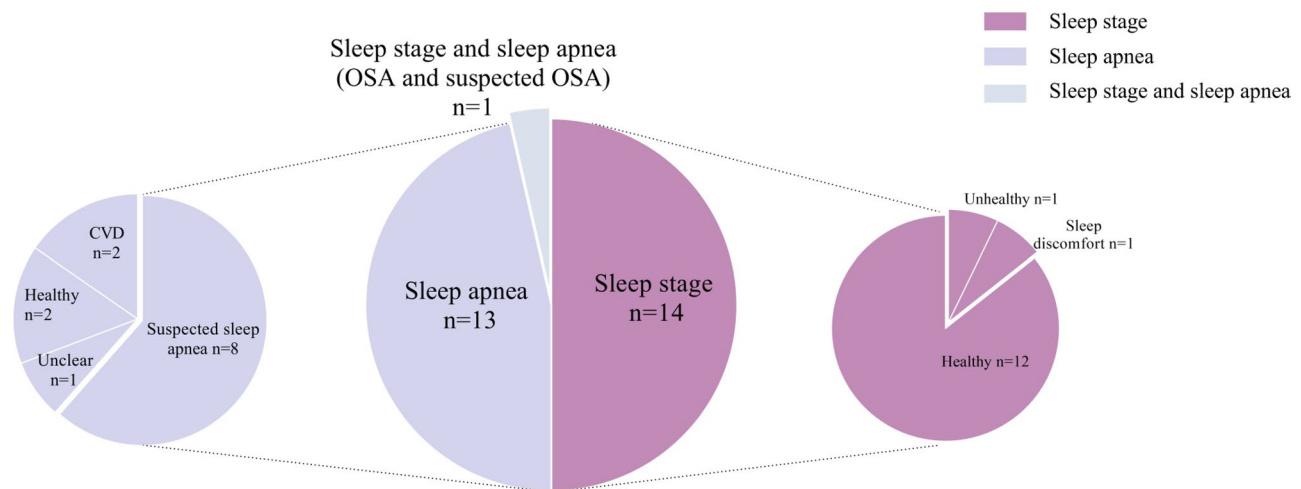


Fig. 2 Distribution of studies and included participants based on health status. The figure presents the distribution of studies and relative participants included in the reviewed studies based on their health status. Abbreviations: n=the number of study, OSA=obstructive sleep apnea, CVD=cardiovascular disease



Fig. 3 Types of finger-worn devices and sensors. Abbreviations: ACC=accelerometer, PPG=photoplethysmography, EDA=electrodermal activity, N/A=not applicable

ranged widely from 0.85 to 1.00. At the moderate threshold (AHI=15), sensitivity ranged from 0.65 to 0.97, and at the severe threshold (AHI=30), it ranged from 0.22 to 0.91. In contrast, specificity showed increase at the cutoff of 5, 15, and 30 events/hour, ranging from 0.03

to 1.00, 0.69-1.00, and 0.91-1.00 respectively. Belun ring demonstrated the highest kappa value at the threshold of 15 ($\kappa=0.79$) and 30 ($\kappa=0.93$). At the cutoff of 5, the RingConn smart ring performed highest kappa value ($\kappa=0.65$).

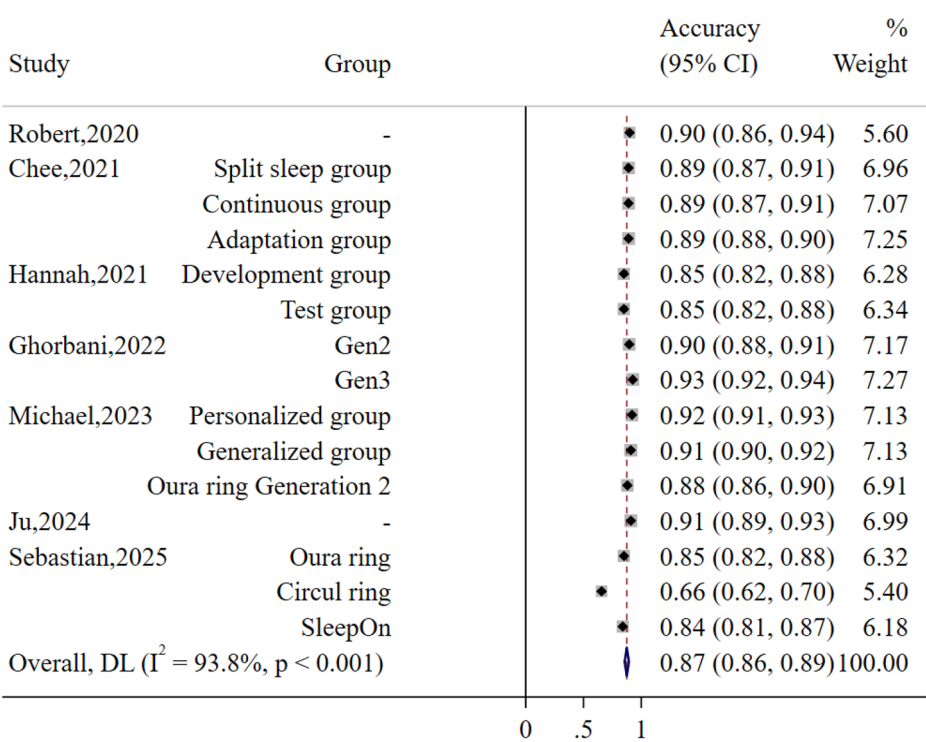


Fig. 4 The accuracy of finger-worn devices for two sleep stages versus PSG. Note: Weights are from random-effects model. Mean and 95% confidence intervals are displayed. Abbreviations: DL=DerSimonian-Laird

Heterogeneity across studies

To explore the sources of heterogeneity, we conducted a subgroup analysis based on different groups (Supplementary Tables 8 and 9). For sleep stage, accuracy was higher with 30s epochs for two-stage sleep (0.88, 95% CI: 0.86–0.90, $I^2 = 94.4\%$) and light sleep (0.66, 95% CI: 0.56–0.75, $I^2 = 99.6\%$). Participants aged 18–50 years had high accuracy across most sleep stages, whereas those >50 years showed low accuracy for two sleep stage (0.78, 95% CI: 0.68–0.89, $I^2 = 96.9\%$). Furthermore, healthy participants exhibited higher accuracy (0.89, 95% CI: 0.88–0.91, $I^2 = 83.9\%$) compared to unhealthy individuals (0.78, 95% CI: 0.68–0.89; $I^2 = 96.9\%$) in two sleep stage. For OSA detection, AHI threshold indicated the accuracy of 0.78, 0.85, and 0.91 for the threshold of 5, 15, and 30.

Bias assessment

As summarized in Supplementary Fig. 5, three (11%) studies demonstrated a high risk of bias in patient selection domain, while 21 (75%) studies were rated as having an unclear risk for this domain due to insufficient details on whether participants were enrolled consecutively or randomly. For the index test domain and reference standard domain, only one (4%) and two (7%) studies demonstrated a high risk of bias. In the flow and timing domain, 20 studies (71%) exhibited a high risk of bias, primarily due to participant exclusions (e.g., insufficient

sleep duration or device malfunctions). Overall, only one study was judged to have a low risk of bias, while 27 were judged to have some risk of bias. To provide a comprehensive review of wearable ring technologies, studies with identified risks of bias were retained.

Discussion

Key findings and technological implications

Wearable technologies have expanded to include body-, wrist-, and finger-worn devices [37]. Finger-worn devices represent an innovative category within this field, offering the ability to monitor diverse sleep metrics, yet their accuracy in staging sleep and detecting apnea remains poorly quantified. This review systematically evaluates their potential for these two tasks.

Sleep stage classification

The American Academy of Sleep Medicine (AASM) calls for rigorous validation of consumer sleep technologies, yet no standardized protocol exists for wearables lacking EEG. Menghini et al. proposed a framework that distinguishes two assessment tiers: macro-level sleep metrics (total sleep time, sleep onset latency, wake after sleep onset, and sleep efficiency) and epoch-by-epoch agreement (accuracy, sensitivity, and specificity) [38]. To maximize information on classifier performance, only studies reporting epoch-level comparisons were included

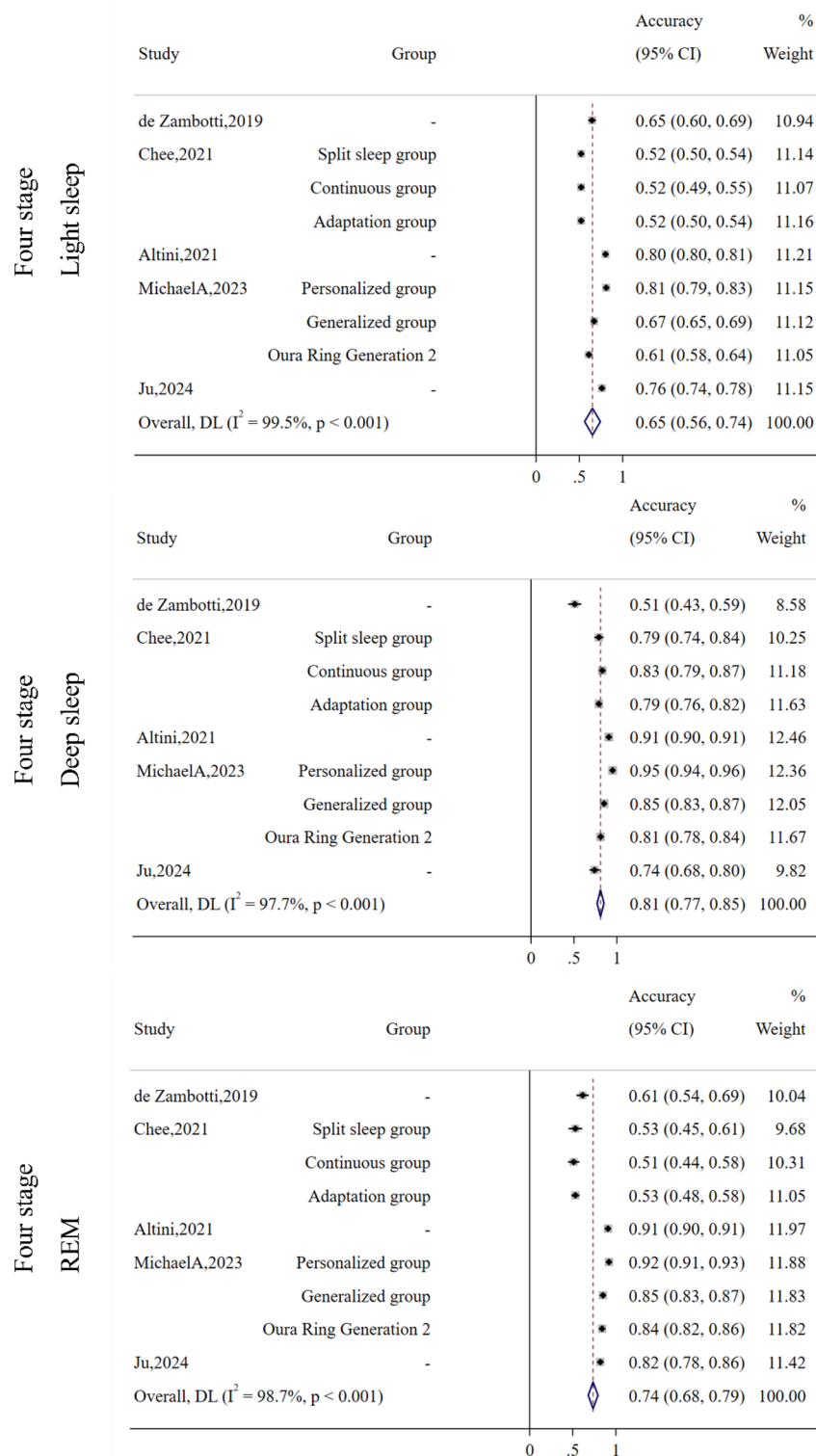


Fig. 5 The accuracy of finger-worn devices for four sleep stages versus PSG. Note: Weights are from random-effects model. Mean and 95% confidence intervals are displayed. Abbreviations: DL=DerSimonian-Laird

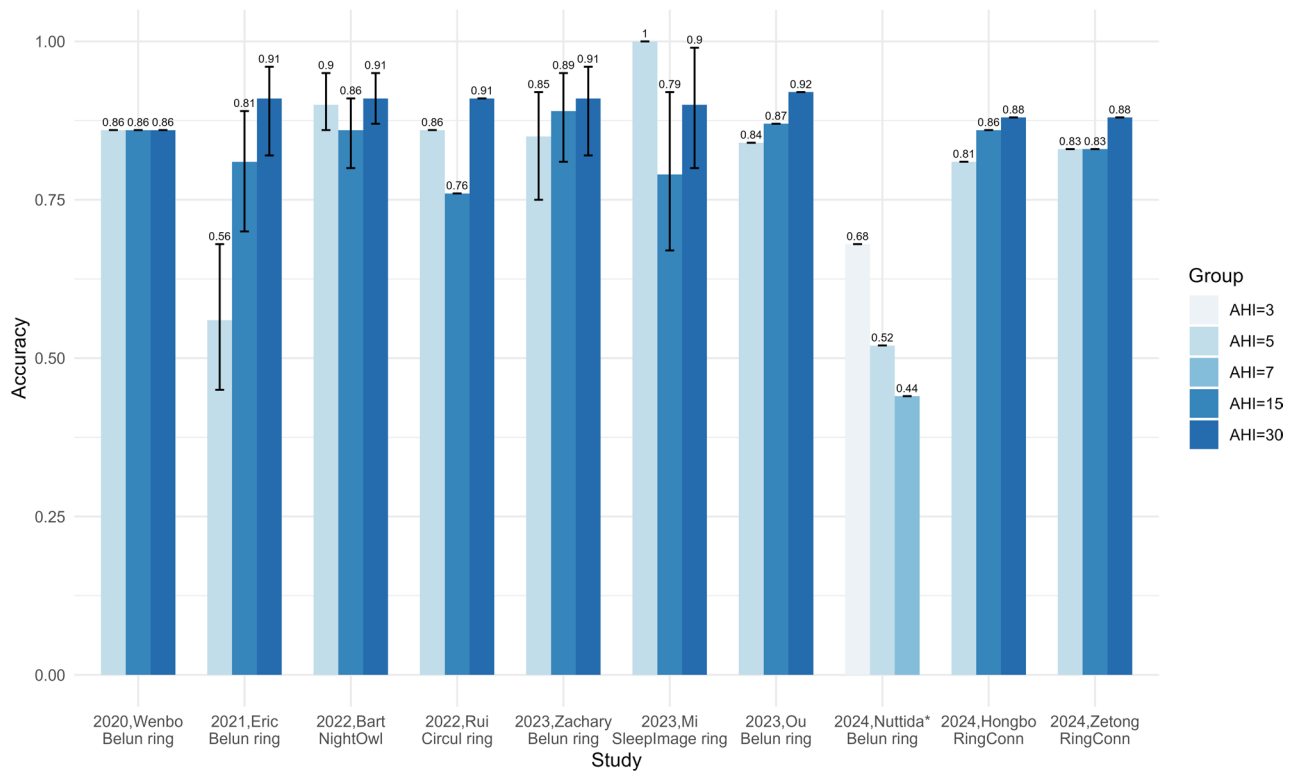


Fig. 6 The accuracy of finger-worn devices for sleep apnea versus PSG. Note: Error bars indicate 95% confidence intervals (95% CI). *Study participants were children. Abbreviations: AHI=apnea hypopnea index

in this review. Sleep stages are derived from physiology either by expert scoring or machine-learning algorithms [39]. Accurate staging is the cornerstone of clinical and research sleep measurement. Finger-worn sensors now reach high two-class (wake vs. sleep) accuracy, but aligning with the AASM four-class scheme (N1, N2, N3, REM) remains unresolved, such findings consistent with a recent review of 62 wearable devices [5]. This four-stage classification more accurately reflects human physiological patterns, thereby necessitating a high degree of precision from finger-worn devices.

Sleep apnea detection

Although prior reviews have examined portable devices for OSA diagnosis, evidence on finger-worn devices is scarce [40, 41]. Our study indicates that smart rings perform adequate sensitivity for mild but high specificity and accuracy for severe OSA, echoing a recent review without meta-analysis [42]. Performance must therefore be interpreted by intended use. High-specificity devices (e.g., Belun Ring at $\text{AHI} \geq 15/\geq 30$) may suited for triaging moderate-to-severe OSA, while high-sensitivity devices (at $\text{AHI} \geq 5$) could enable initial screening in primary care or at home, with referrals for PSG confirmation.

Notably, devices such as the Belun ring demonstrate concurrent monitoring capabilities for both sleep staging

and OSA. Its second-generation algorithm, trained on first-generation datasets, integrates dual proprietary neural networks dedicated to sleep stage classification and respiratory event detection [36]. This multi-functionality advances holistic sleep assessment through integrated physiological capture.

Methodological and population considerations

This systematic review identified accelerated research growth, with 70% of relevant studies published within the past quadrennium. We evaluated 11 finger-worn devices, among which Oura Ring and Belun Ring constitute the most extensively researched. Notably, technological evolution has shifted from single-sensor to multi-sensor configurations [8], predominantly integrating ACC and PPG to capture concurrent physiological, kinematic, and health parameters [43]. Contemporary iterations further incorporate temperature sensors, enabling comprehensive monitoring of sleep-associated physiological dynamics, which align with the requirements of core sleep medicine biomarkers.

Finger-worn devices enable cost-effective multi-night sleep monitoring in natural environments. Continuous longitudinal data acquisition mitigates single-night sampling bias while tracking dynamic trends in physiological and environmental covariates. This approach enhances

clinical decision-making through pattern recognition of interrelated variables, improving ecological validity in real-world contexts [44, 45].

Despite prior validation of finger-worn devices' sleep monitoring potential [21, 46], current evidence reveals inconsistent PSG-referenced accuracy. Most devices have shown improved accuracy over time, primarily due to algorithmic optimizations and integration of multi-sensor technologies. Notably, the lack of transparency regarding proprietary algorithms-functioning as black boxes, which obscures the specific mechanisms. This opacity also restricts end-users and clinicians from accessing raw data, limiting further validation and refinement of the technology [5]. Furthermore, a considerable number of studies fail to focus on the imbalanced datasets in sleep-stage classification, limiting a comprehensive assessment of device performance.

Methodological heterogeneity in algorithms and sensor configurations likely underlies inter-study discrepancies. For instance, personalized algorithms (e.g., Happy Ring [22]) demonstrate superior accuracy to generalized approaches, suggesting customization as a critical development pathway. Epoch length is also a significant source of heterogeneity and influences accuracy estimates. Using the standard 30-second epochs yielded higher pooled accuracy for two-stage and light sleep staging. However, this subgroup also exhibited high statistical heterogeneity, suggesting that even among studies using the same epoch length, other factors (device, population) cause substantial performance variation. In addition, population heterogeneity appears to be another contributing factor, as evidenced by studies reporting variable accuracy results for the same finger-worn device, although hand dominance shows negligible effect [25].

Limitations

This review systematically evaluates finger-worn devices against PSG standards for sleep staging and OSA diagnosis, providing critical developmental insights. However, several limitations should be acknowledged in this review. The primary limitation of this review is the scarcity of published validation studies for the available consumer devices. To address this, we sought additional studies by checking references or through snowball sampling. Despite these efforts, it is possible that some finger-worn devices were overlooked. Second, the strength of evidence is tempered by the methodological quality of the included studies. Most studies were rated as having high or unclear risk of bias, which reflects the current lack of standardized protocols and reporting frameworks in this rapidly evolving field. This may inflate or bias the reported accuracy estimates. Third, as ring technology (including sensor hardware, firmware, and proprietary algorithms), evolves rapidly, the performance reported

in the current literature cannot be directly generalized to future versions or substantially revised versions. Another limitation is that costs were not extensively addressed. Prices and business models vary widely and fluctuate with market dynamics, making cost considerations important for prospective users. Future studies should therefore assess the cost-effectiveness and add value of these wearable and nearable technologies. Finally, the limited applicability of current findings to the general population stems from spectrum bias. For sleep staging, the predominant enrollment of healthy adults (to minimize comorbidity confounders) restricts generalizability to pathological populations, as the performance of finger-worn devices in patients with diverse sleep pathologies remains largely unexamined. Conversely, studies on OSA have largely focused on high-risk, clinically suspected populations, typically overweight adults over 50 years old with moderate to severe apnea. This leaves individuals with mild OSA or those with comorbid condition (e.g., cardiovascular or neurological diseases) underrepresented. A major evidence gap also exists for pediatric and adolescent populations, where finger-worn devices have been scarcely evaluated. These limitations constrain the broader applicability of our findings and highlight the need for future research in more diverse clinical and demographic cohorts.

Clinical relevance and future directions

Our study has significant clinical relevance. Current devices demonstrate proficiency in detecting severe OSA and differentiating basic sleep/wake states; however, their accuracy in identifying mild OSA and distinguishing the four sleep stages remains inadequate for clinical diagnosis. Furthermore, the applicability of findings is often limited by population homogeneity, which may not reflect the complexity encountered in diverse patient profiles.

Future research should prioritize several key directions:

1. Promoting open-source frameworks: The current dominance of proprietary algorithms (8/11 devices in review) hinders independent validation and innovation. Open-source approaches would accelerate progress in these areas.
2. Addressing pediatric/adolescent populations: Current evidence is scarce, with only three studies in this review involving children or adolescents. Developing age-appropriate sensor placements and algorithms is an urgent need.
3. Establishing methodological consensus: Standardization is critical for quality control (QC) steps of data processing, algorithm calibration, epoch length, and feature definitions across studies and devices.

4. Fostering interdisciplinary collaboration: Advancing finger-worn device technology requires close, sustained collaboration among medical, engineering, and other relevant disciplines to foster innovation and expand clinical applications.

Conclusion

Finger-worn devices may be used for longitudinal sleep/wake tracking in healthy adults or as a triage tool for severe OSA, but should not replace PSG for diagnosis of mild OSA or detailed sleep architecture assessment. Achieving PSG-comparable accuracy in multi-stage sleep analysis and mild OSA detection requires addressing algorithmic transparency, population diversity, and methodological standardization. Therefore, future innovation should prioritize clinically actionable insights through interdisciplinary collaboration.

Abbreviations

ACC Accelerometer
PPG Photoplethysmography
NTC Negative temperature coefficient
IBI Interbeat interval
SD Standard deviation
HRV Heart rate variability
HR Heart rate
SpO₂ Oxygen saturation
PR Pulse rate
EDA Electrodermal activity
REM Rapid eye movement
OSA Obstructive sleep apnea
PR Pulse rate
PAT Peripheral arterial tone
ODM Overnight digital monitoring
ODI Oxygen desaturation index
CVD Cardiovascular disease
DL DerSimonian-Laird
AHI Apnea hypopnea index
N/A Not applicable

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12967-026-08214-x>.

Supplementary Material 1

Acknowledgements

We thank all the investigators of the studies included in this systematic review and meta-analysis.

Author contributions

YXJ, JWX, HJY, YZH, WJM, XMF, HMW, LX and JW have substantial contributions to conception and design, acquisition of funding, data and interpretation of data; YXJ and JWX analyzed the data and drafted the article, HMW, LX and JW revised it critically for important intellectual content. YXJ, JWX and HJY contributed equally to this work. JW is the guarantors. All authors contributed to final approval of the paper.

Funding

This work was funded by the Natural Science Foundation of Guangdong (2025A1515011659) and the fundamental research funds for the central universities, Sun yat-sen university (24qnpy211).

Data availability

All data generated or analysed during this study are included in this published article and its supplementary information files.

Declarations

Ethical approval

Ethical approval is not required because this study is retrieved and synthesized data from already published studies.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

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Received: 19 November 2025 / Accepted: 23 April 2026

Published online: 15 May 2026

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